Swinburne University of Technology

COS30049

Computing Technology Innovation Project

Assignment 1

Group 4

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Part A: Introduction & Team Overview

1.1 Project Background & Overview

The Swinburne University of Technology offers the Computing Technology Innovation Project (COS30049) course, providing students with a collaborative environment for developing innovative ideas applicable to real-world contexts. Throughout the course, students are learnt to integrate project management lifecycles, artificial intelligence (AI), and web development techniques to address complex challenges on a global scale.under lecturer's recommendations and guidance.

Due to the rising sophistication of cycber-attacks and overwhelming fake digital data sent to Internet users, my team decided to launch a potential web-based platform capable of detecting both Spam and Malware mail and intelligently categorize digital news content by implementing suitable machine learning models. Furthermore, the platform serves as a dynamic tool integrated to protect users from malicious communication while simultaneously organizing information into meaningful categories. Moreover, the interesting approach allows team members to gain hands-on experience in executing a full-scale cybersecurity project from inception to deployment.

1.1.1 Problem Statement

In today's world, the rocket advancements in man-made intelligence enable humans to complete complex tasks in the workforce and academic environments more efficiently and quickly, supporting in better-informed decisions and improved productivity. However, these technological innovations have also given rise to unexpected challenges. As reported by the World Economic Forum's Global Risks in 2025, misinformation and disinformation have been classified as the leading global risk for the second consecutive year, driven by a significant surge in AI-generated "fake news" that erodes social trust. Specificially, spam and malware emails have escalated significantly in recent years, becoming one of the most widespread cybercrime issues. Additionally, the number of phishing emails sent daily is approximately 3.4 billion, accounting for roughly 1.2% of all global email traffic (Keepnet Labs, 2026).

In Vietnam, the volume of network traffic accessed has risen rapidly, contributing to higher levels of online activity and cybersecurity threats. In the previous year, as reported by Internet Society Pulse, nearly 78% of Internet users spent time on communication, learning, and commerce, generating fertile virtual ground for cyberattacks. Currently, cybersecurity threats have continued grown rapidly in scale and sophistication. In the third quarter of preivous year alone, authorities recorded over 6.5 million personal accounts were stolen, including around 4,000 phishing domains and roughly 547,000 DDoS attacks were created by automated and AI-assisted methods. During researching and anaylzing process, Vietnam experienced roughly 29,000 ransomware attacks on businesses and widespread malware infections in the year 2024 that highlights the vulnerability of corporate and personal systems, according to reports from Lao Dong News. Consequently, these trends indicate the urgent need for intelligence. Consequently, my team determined to publish an intelligent website designed to analyze, detect, and mitigate misinformation, phishing, and other email-based cyber threats, enhancing digital security and user trust.

Beyond 2025, the global digital landscape, and Vietnam in particular, is expected to face an increasingly intense "AI race." In addition, experts predict that in the foreseeable future, over 90% of online content could be synthetically generated or AI-enhanced, making it nearly impossible for humans to distinguish between legitimate news reports and deepfakes (World Economic Forum, 2025). Surprisingly, the existence of "Agentic AI" is expected to drive a 40% rise in automated phishing attacks, as malicious bots gain the ability to engage victims through real-time and highly personalized interactions (Google Cloud, 2025). As noted by Gartner, the financial impact of misinformation is projected to exceed $30 billion globally, pushing organizations to adopt flexible and automated verification tools as a standard security measure in 2028.

1.1.2 Proposed Solution

The proposed solution represents a comprehensive analytical platform designed to address the dual-threat landscape of modern digital communication. By implementing Optical Character Recognition (OCR) and powerful machine learning models within a FastAPI ecosystem, the interactive website facilitates a robust and end-to-end data processing pipeline. As a result, the architecture is engineered to couple with raw data acquisition from core analytical logic, allowing for a highly scalable and modular approach to content verification. In detail, the framework has three primary functional layers:

- Interactive Presentation Layer: The backend progresses uses React coding platform to simplify the backend's complex data, serving as the final link between the technical analysis and the user, converting probability scores into clear, actionable results that make advanced digital security practical for everyday use.
- Hybrid Data Coleection Layer: In this secured layer, the platform implements an adaptive input handler that serves as the gateway for all incoming data. by utilizing cutting-edge techniques, the website can extract textual information from visual media, including screenshots and flyers, ensuring the underlying models receive clean and structured data regardless of the initial input format.
- Modular Analytical Engine: At the heart of the platform is a dynamic routing mechanism. Based on user requirements, the system directs processed text to specialized neural networks or machine learning models. in this stage, the model ensures that Spam/Malware detection and Newspaper categorization are treated as distinct cryptographic and linguistic challenges, maximizing the precision of each output.

1.1.3 Project Objectives

The primary goals of the fascinating project are to develop a reliable and user-friendly assistant that empowers Internet users to navigate efficiently. Besides that, the team leader publishes the following project's objectives to ensure the system meets high technical and functional standards:

- Implementation of High-Precision Models: The website aims to maintain classification accuracy exceeding 90% for both spam detection and news labeling. Consequently, the high level of precision

is extremelycrucial for building user confidence and ensuring the website works smoothly

- Robust Visual Intelligence: Team members choose the OCR module, which maintains high character recognition rates across multiple image formats, allowing them to "read" screenshots and types as effectively as raw text.
- System Responsiveness: By optimizing the backend data pipeline, our team selects to use FastAPI, ensuring high-speed and asynchronous processing of user requests.
- Accessibility through Design: The platform simplifies complex data for the end-user by using a React-based dashboard to convert intricate probability metrics into intuitive, color-coded visual indicators. As a result, the website showcases advanced data science, ensuring that technical findings are immediately understandable and actionable for non-expert users.
- Scalable Architecture: Design a modular system that easily incorporates and updates machine learning models or security filters in the future as digital threats continue to evolve.

1.2. Team Introduction

A group of aspiring students initiated the project with a shared passion for creating secure digital environments. Additioanlly, each individual has unique strengths which ensure every layer of our project—from design to data protection—is handled with expertise. Thus, every member plays a vital role in ensuring that the project not only meets academic standards but also addresses real-world challenges in digital safety.

1.2.1 Team Members, Roles, and Responsibilities

Name	Student ID	Roles	Responsibility
Truong Ngoc Gia Hieu	SWS01217	Team Leader	- Defining project vision and setting deadlines. - Managing and delegating team tasks - Facilitating discussions and resolving conflicts - Leading brainstorming for innovative solutions
Pham Dinh Duc Anh	SWS01484	Vice Leader	- Supporting leader in backend development. - Training machine learning modules. - Ensuring logical and maintainable model - Debugging and resolving technical issues.
Duong Nguyen Dang	SWS01389	Member	- Designing responsive website interface. - Collaborating on functional feature

			analysis. - Optimizing for best user experience (UI/UX).
Pahan Partirathna	SWS01652	Member	- Assisting with design alignment. - Conducting code reviews. - Supporting development progress to meet deadlines.

Figure 1: Team Introduction

1.2.2 Technical SWOT Analysis

As reported by Gartner in the year 2025, using the SWOT framework ensures we are prepared for external threats, such as the predicted $30 billion enterprise spend required to battle misinformation by 2028. By utilizing SWOT techniques, our team can identify our Strengths in specialized technical expertise, address Weaknesses in existing detection protocols, and determine Opportunities to implement powerful tools to complete the project. Furthermore, the technique not only allows us to break invisible barriers during the workflow but also remains resilient against the rising tide of AI-driven narrative attacks and deepfakes.

- Strengths: by implementing advanced technologies, including the Optical Character Recognition (OCR) and Machine Learning model, the platform showcases sophisticated processing capabilities for both textual and visual data. For the front-end part, our team utilizes React and the Node.js framework to develop a highly responsive and scalable user interface, ensuring a seamless user experience for non-technical users.
- Weaknesses: The primary limitation is the AI model's sensitivity to data quality, where unstructured, biased, and noisy datasets can significantly impact the training phase and overall performance. Current operational gaps include OCR instability with complex image layouts and a lack of data persistence, limiting the system's ability to leverage historical data for future model optimization
- Opportunities: Currently, the rising need for better security generates a great opportunity for us to expand. By initiating and executing this fascinating project, this gives us a great chance to sharpen our professional skills and gain hands-on experience with cutting-edge tools, ensuring we are prepared to tackle the next generation of digital threats
- Threats: My team faces several challenges from the rapid evolution of AI-generated phishing and strict regulatory constraints. Furthermore, competing with existing commercial solutions requires us to constantly innovate to maintain a competitive edge.

Part B: Project Management & Design

2.1. Project Requirement Analysis

The project requirement analysis concentrates on addressing real-world challenges with innovative projects by utilizing powerful tools and applications. Notably, our team decided to publish the fascinating project to solve the assignment's requirements by initiating a structured management plan, featuring a clear timeline and a prototype design. Consequently, the potential website ensures that the machine learning model is not only highly-structured and logical but also user-friendly and easy to navigate.

2.2. Project Management Plan

In the report, this section features a completed project management plan to ensure maximized productivity, a clear vision, and effective collaboration among team members, leading to the seamless integration of the machine learning model into the final web deployment. Besides that, the plan serves as a roadmap to keep track of key milestones, manage technical resources, and ensure that every phase, from initiation to closing, is executed on schedule and within the project's scope.

2.2.1 Phase I: Initiation

The TruthGuard website was initiated and launched to bridge the gap between technical cybersecurity and content integrity, which was driven by our slogan "Verify, Protect, Trust". While project decisions and technical directions are decided that align with assignment requirements and industry standards by our team, the platform is developed under helpful advice and potential recommendations of Mr.Hoang (Arthur) Nguyen. Additionally, the final goal of the project aims to deliver an intelligent system that effectively fights against malware and spam mail while providing a secure and reliable tool for verifying newspaper authenticity.

From a leader's perspective, creating clear boundaries is essential to ensure the technical combination of the system. Furthermore, our team has established a robust and trusty framework for accountability and transparent decision-making by conducting several meetings and documenting main points during all meetings during the process. Consequently, the innovative solution allows us to prioritize high-quality and clean code over unnecessary complexity, which makes sure the system is maintainable. The project is defined as follows:

- In-scope:

    - Implementing a centralized web interface combining security scanning and newspaper certification.
    - Researching and analyzing suitable machine learning models for training and classification.
    - Creating a clean and maintainable Python-based backend to ensure system transparency.

- Out-of-scope:

- Implementing a centralized web interface combining security scanning and newspaper certification.
- Researching and analyzing suitable machine learning models for training and classification.
- Creating a clean and maintainable Python-based backend to ensure system transparency.

Thus, my team believes that establishing core objectives and scope offered a comprehensive roadmap for the project. Next, the following section details our strategic planning to ensure time delivery and technical excellence.

2.2.2 Phase II: Planning

In this part of the report, we outline the logical and strategic framework that will govern the project's development, ensuring the project works on the right track and technical execution remains aligned with the defined scope and timeline.

Based on Swinburne's material, Work Breakdown Structure (WBS) is a hierarchical decomposition of the total scope of work to accomplish the project objectives and create the deliverables. Therefore, a well-defined WBS improves project planning accuracy by providing a structured basis for cost estimation, scheduling, and resource allocation (Kerzner, 2022). Additionally, the figure below highlights the detailed WBS for the TruthGuard website, providing a clear and visible roadmap for resource allocation and scheduling based on our primary deliverables.

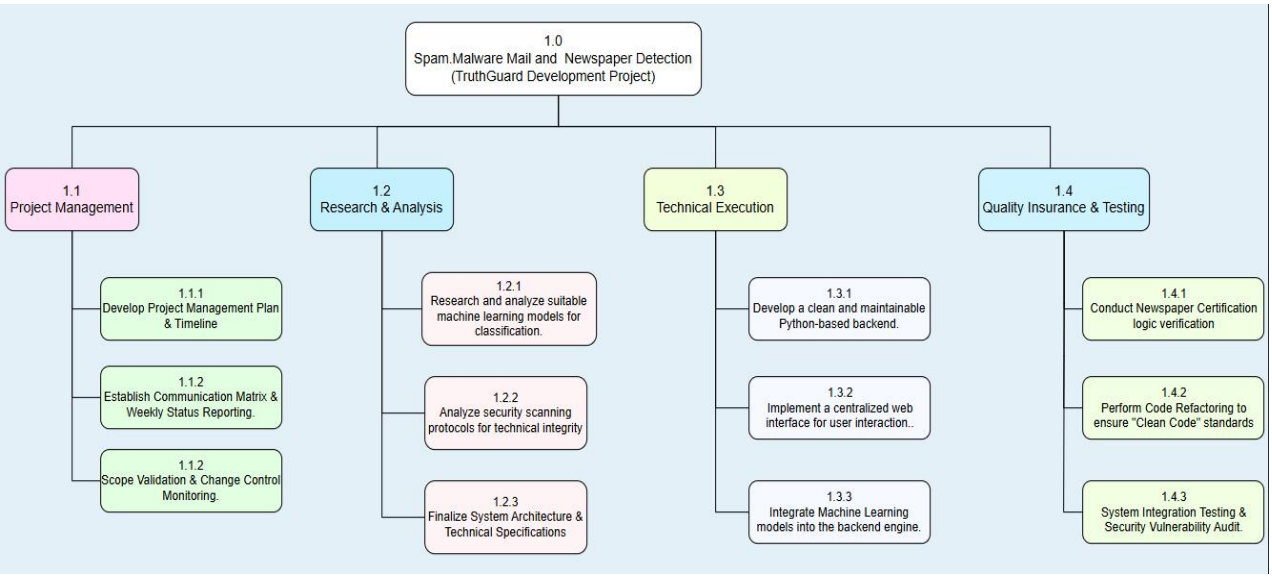

Figure 2: TruthGuard Work Breakdown Structure

The following Gantt chart showcases the timeline for the TruthGuard platform over a 13-week duration in the course. According to Gantt (1919), visual scheduling techniques enhance managerial control by clearly illustrating task sequences, durations, and progress. Therefore, the Gantt chart not only provides a clear roadmap in development phases - includig monitor milestones - but also ensures the timely and systematic implementation of the TruthGuard website.

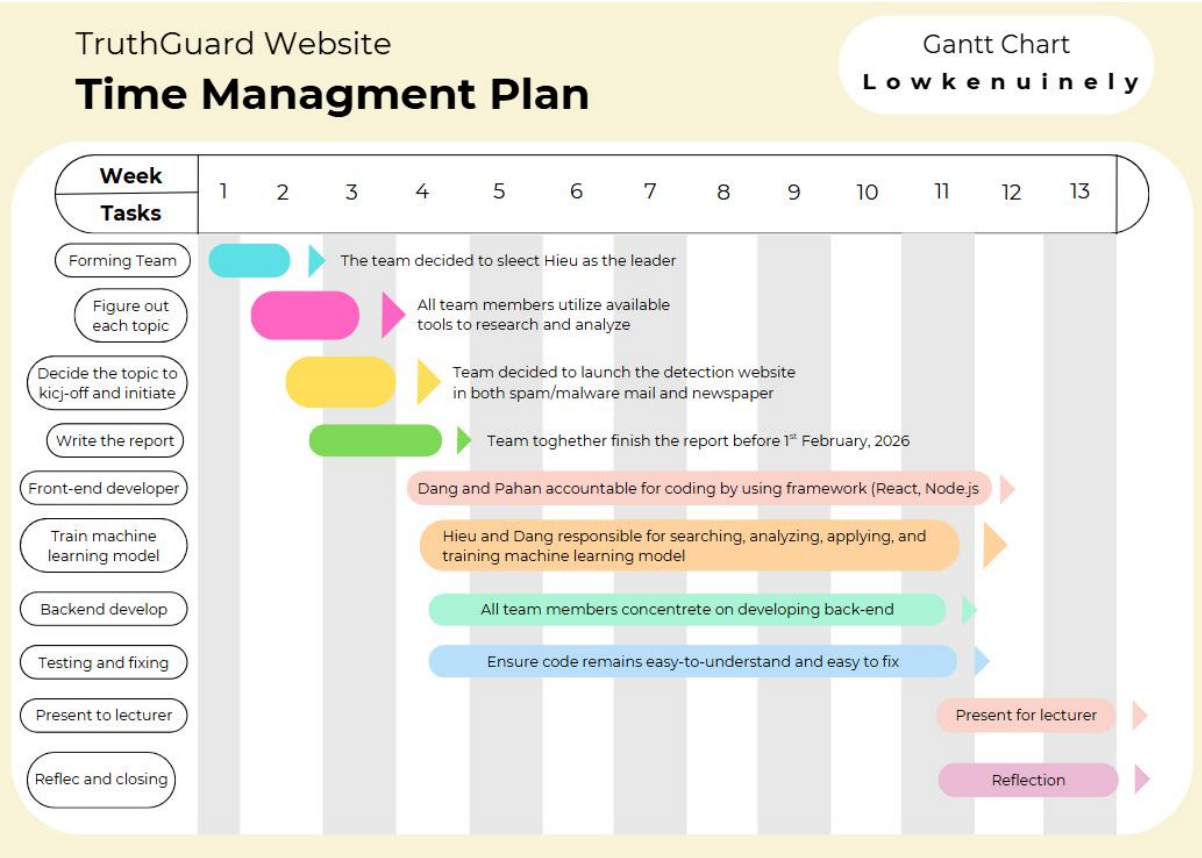

Figure 3: Gantt chart

Beyond tracking the general process, the Gantt chart serves as a vital monitoring tool for this project, ensuring the team remains aligned with key academic milestones. Notably, the chart acts as a visual reminder of critical deadlines, allowing us to synchronize our technical development with the following course requirements:

- Milestone 1 - Assignment 1 (Week 4): Comprehensive Project Management Plan, establishing the foundation for our innovation project.
- Milestone 2 - Assignment 2 (Week 8): Intermediate Progress Report, focusing on back-end development, including the training of machine learning models and source code integrity.
- Milestone 3 - Assignment 3 (Week 12): Final Project Delivery, encompassing the fully functional website and a comprehensive final report

Besides that. Our team selects available tools, including Google Docs, GitHub, and Google Drive, and each serves a specific purpose in the workflow:

- Google Docs: Utilizing for real-time collaborative documenting, offering team members the ability to simultaneously draft reports, refine project requirements, and maintain consistency in our academic submissions.

- GitHub: Operating as the central repository that contains all source code. Therefore, GitHub allows us to track changes, review code contributions, and maintain technical transparency throughout the development of the TruthGuard platform.
- Google Drive: Functioning as a secondary centralized hub for backup and file storage, ensuring all project resources, materials, and administrative files are securely stored and accessible to every team member.

Additionally, as stated by Peter F. Drucker, “the best way to predict the future is to create it,” which highlights the necessity of conducting an initial risk assessment for effective management in uncertain environments. Because of the above reason, creating an Initial Risk Assessment figures out invisible barriers, helps the team to develop proactive mitigation strategies based on the WBS before moving into the Execution phase.

Risk Category	Potential Risk	Impact	Mitigation Strategy
Technical	The machine learning model yields low accuracy in the expected outcome	High	Start the "Research and analyze suitable models" task (WBS 1.2.1) early and utilize diverse datasets for the training step.
Integration	Conflict issue between the Python backend development and React and Node.js frameworks during the coding phase.	High	Adhere to the "Integrate Machine Learning models into the backend engine" (WBS 1.3.3) schedule and perform continuous integration testing.
Time	Tight academic deadlines for three assignments during the course, leading to rushed development.	Medium	Strictly follow the Gantt chart milestones and conduct weekly status reporting (WBS 1.1.2) to monitor progress.
Quality	Source code becomes complex and difficult to maintain (technical issue).	High	Ensuring 'Clean Code' standards (WBS 1.4.2) are a mandatory step in every sprint.
Scope	Scope creep due to over-complicating the features of the detection website.	Medium	Strictly follow the "Scope Definition & Change Control" (WBS 1.1.3) protocol to prioritize core functionalities.

Figure 4: Initial Risk Assessment

By generating comprehensive potential project planning tools, including Initial Risk Assessment, WBS, and Gantt chart, our team has secured a solid foundation to officially transition from the Planning step to the Execution phase, focusing on the technical implementation of the TruthGuard platform

2.2.3 Phase III: Execution

At the current stage, our team is actively moving from the design phase to the development phase. While the detailed UI/UX interfaces are showcased in the Project Design section, the Execution phase focuses on the following activities:

- Translating Design into Code: Utilizing the Figma sketches as a blueprint to develop the functional interface with React and Node.js.
- Back-end and machine learning development: Implementing the core logic and training machine learning models for accurate detection.
- Code Quality Control: Ensuring all source code remains 'easy to understand' and well-documented as we build the platform

After determining essential technical components during development, our team continues to dive into the fourth phase: Monitor and Control, focusing on tracing our team's progress based on the Gantt chart and managing potential risks to guarantee the project stays on the right track.

2.2.4 Phase IV: Monitoring & Controlling

In the fourth phase, the project leader serves a pivotal role in maintaining the project's track by comparing current progress against the baseline, managing changes, and ensuring the team adheres to the 'Clean Code' standards established in previous phases.

As noted by Stephen Covey, "Even if you are on the right track, you will get run over if you just sit there," highlighting the significance of generating a Status Tracking Table to monitor task progress, support task management, and enable team members to make proactive decisions. By using Status Tracking Table, our team can secure that project tasks and deadlines align with the detailed plan, avoid potential risks, and promptly implement corrective actions to achieve the final expected product. The following Status Tracking Table showcases the current progress of the TruthGuard project, providing a clear comparison between planned milestones and actual outcomes.

Status Tracking Table

Tasks	People	Status	Files	Start Date	End Date	Note
Sketch the UI/UX Interface	Dang, Pahan	Done	Tệp	22 thg 1, 2026	24 thg 1, 2026	
Research and Analyze suitable model	Hieu, Duc ...	In Progress	Tệp	24 thg 1, 2026	27 thg 1, 2026	
Sequence Diagram descrition	Duc ...	In Progress	Tệp	24 thg 1, 2026	26 thg 1, 2026	
Write description for UI/.UX Design	Dang, Pahan	Done		22 thg 1, 2026	22 thg 1, 2026	

Figure 5: Status Tracking Table

Based on the Swinburne lecture in the third week, quality control is a critical part of the Monitoring and Control phase, ensuring final project deliverables meet the defined acceptance criteria, expected outcomes, and quality standards. Furthermore, according to Deming (n.d.), quality is everyone's responsibility, emphasizing that effective quality control requires collective commitment across all levels of an organization rather than reliance on a single department. For maintaining the integrity of the TruthGuard website, our team leader decided to generate quality control measures that guarantee every system component is thoroughly validated before integration.

- Peer Code Reviews: Every code implementation during the workflow is cross-checked to ensure it aligns with our core objective: "Easy to understand and easy to fix". Besides that, the first quality control principle ensures that the logic remains transparent and the documentation is clear, showcasing a solid foundation for rapid debugging and ensuring the final deliverables meet the team's expected outcomes.
- Design-to-Code Validation: Weekly evaluations are conducted by our team leader during the project to verify that both Back-end and Front-end development are progressing well. Notably, the second quality control factor ensures that functional components accurately reflect the Figma blueprints and technical specification. Therefore, the leader guarantees visual consistency and system integrity that prevents design drift by creating alignments and following the roadmap while ensuring the final product meets all predefined quality standards.
- Logical Flow: By creating a sequence diagram via the draw.io website, our leader verifies that the interaction between the Front-end, Back-end, and Database is architecturally sound. Thus, this visualization ensures that the system logic remains transparent and "easy to fix", offering opportunities for the team to identify and resolve potential integration bottlenecks before they impact the final delivery.

According to the Project Management Institute (PMI), establishing a risk management plan is an important part of systematically identifying, analyzing, and responding to uncertainties, thereby increasing the likelihood of achieving project objectives. In this step, the project leader decides to generate a risk planning sheet in order to predict invisible barriers and external factors that can affect the project. In detail, the figure below represents a risk management sheet for securing the project's stability and keeping the right track on the deliverables.

RISK PLANNING SHEET			
POTENTIAL RISK	LIKELIHOOD OF OCCURRENCE	IMPACT IF OCCURRED	PREVENTION PLAN / SOLUTION
Technical issue: Code becomes too complicated to fix or updated	Medium	High	Enforce the "Easy to understand" rule via peer reviews and sequence diagrams
Scope Creep: Adding too many features beyond the Figma design.	High	Medium	Conduct weekly evaluations to strictly align development with the initial blueprints.
Integration Failure: Front-end and Back-end don't sync.	Medium	Very High	Use draw.io sequence diagrams to map logic before implementation; perform daily integration tests.
Resource Unavailability: A team member is unable to work.	Low	High	Maintain a transparent Status Tracking Table so others can cross-cover tasks easily.

Figure 6: Risk Planning Sheet

By managing risks based on the planning sheet, controling quality control factors, and keeping right track on the status table, the leader safeguards the project within its defined scope and timeline. Therefore, these management tools ensure our final product meets the project's outcomes while adhering to the 'easy to understand' standard. Consequently, the team has successfully fulfilled the assignment's requirements, allowing a smooth transition to the final phase.

2.2.5 Phase V: Closing

In the final stage, our team focuses on confirming that all deliverables meet the final acceptance criteria and obtaining official stakeholder approval. After presenting the TruthGuard website to the lecturer in the final week, the team gathers to conduct a post-implementation review and document lessons learned during the project.

2.3 Projec Design

In the report. The project design section serves as a significant bridge between conceptual management and technical execution. Therefore, transforming high-level requirements and sketches into a tangible architectural blueprint that secures the TruthGuard website remains robust, scalable, and the code remains easy to read and fix. Consequently, the team with a clear guideline for successful implementation by detailing the logic and interactions between technologies components

2.3.1 Project Rationale & Vision

The project rationale aims the fundamental necessity of TruthGuard which showcase the critical gap in digital trust caused by the proliferation of malware, spam emails, and fake news in the era of globalization. Besdies that, the project vision justifies the transition from theoretical planning to actual technical execution, ensuring the website addresses a real-world demand for a safe, transparent, and verifiable online experience. Additionally, our team concentrates on mitigating these deceptive threats through rigorous data verification and secure architectural design by generating a management plan.

Next, the project vision defines the long-term strategic goal of establishing TruthGuard as a benchmark for the innovative project. Moreover, the vision not only indicates the aspiration to restore user confidence in digital information but also ensures that every technical component remains easy to read and fix for future scalability. Moreover, my team opt for generating detailed strategies in the management part that guarantee the platform is adapted to the evolving digital landscape. Consequently, this approach secures that TruthGuard serves as a sustainable solution, fulfilling its mission to provide a transparent and secure user experience long after the initial development phase.

2.3.2 Sequence Diagram & interaction

According to Ian Sommerville, a sequence diagram represents a standardized way to document system interactions, enabling effective communication among stakeholders and ensuring consistency in system design. Besides that, the sequence diagram below demonstrates system components within the TruthGuard platform and the chronological interactions between them to fulfill a specific user request.

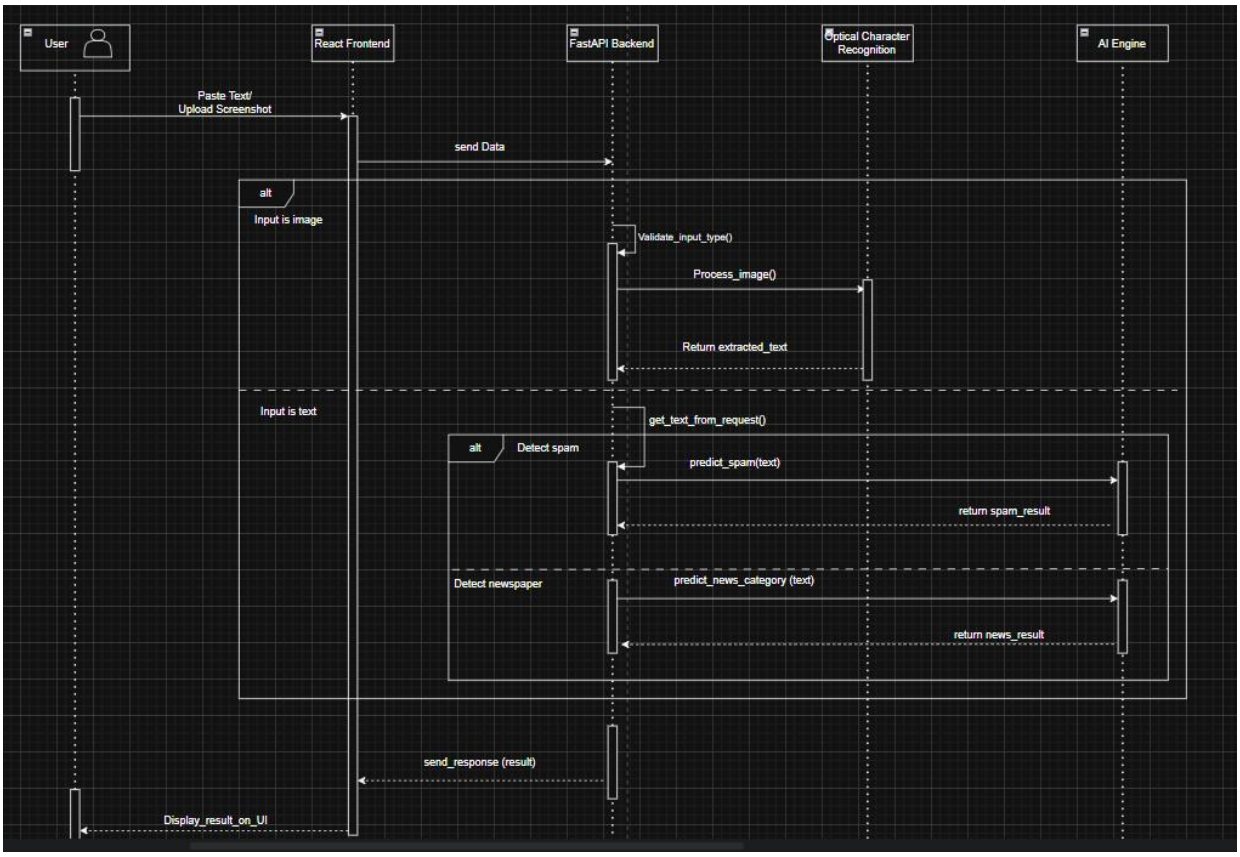

Figure 7: Sequence Diagram

The sequence diagram consists of five steps in the processing workflow, including User, React/Node.js Framework, FastAPI Backend, Optical Character Recognition, and AI Engine. Based on the flow, our team can guarantee a seamless transition from user input to data verification, maintaining a modular architecture that keeps the system robust and the code easy to read and fix. In addition, the following description demonstrates crucial steps on how the website actually works:

- Step 1:- Input Submission: The platform initiates the process by either pasting text or uploading a screenshot to the React/Node.js Framework. Then, the data will be transmitted to the frontend framework.
- Step 2 - Data Validation: FastAPI receives data from user's input and executes a validation function to determine whether the incoming data is an image or raw text, securing the system remains stable.
- Step 3 - Optical Character Recognition (Image Path): If the input is an image, the backend sends a request to the OCR component to extract the text content and return it to the backend.
- Step 4 - AI Engine Detection (Text Path): The FastAPI Backend sends the text (either provided or extracted) to the AI Engine to perform simultaneous checks for spam detection and news categorization.
- Step 5 - Results Showcases: After receiving results from the AI Engine, the FastAPI Backend sends a response back to the React/Node.js Framework, which displays the final verification result on the UI for the User

By following the above workflow, our team offers high levels of control over the system's logic and interactions, ensuring the project meets its quality requirements before closure.

2.3.3 UI/UX Design & User Experience

In this process, the UI/UX design step serves as a sketch that showcases our stunning ideas into a visual structure, ensuring the platform remains a logical processing and user-friendly interface.Based on the drqing, the team can verify that the layout is easy to understand and that the navigation flow aligns with the project's core mission of transparency. As a result, the drawing plays a definitive role in development, which guarantees the final product remains consistent with the initial vision while providing a clear roadmap for implementing a safe and verifiable online experience.

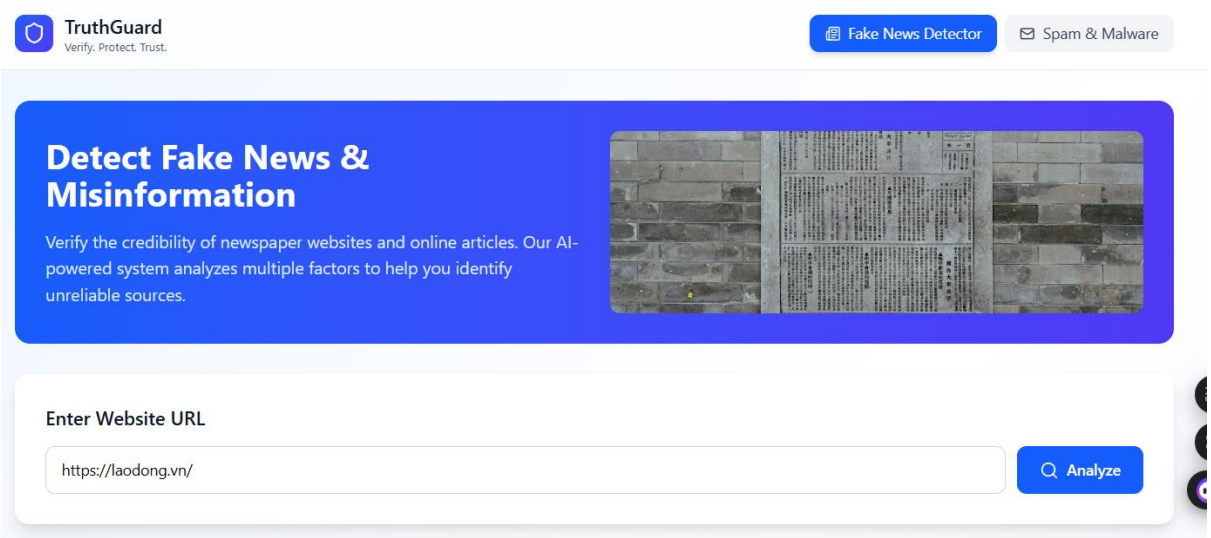

Figure 8: Fake News Detector Interface

First and foremost, our website includes two main functions: Fake News Detector and Spam and Malware Mail filter. As illustrated in Figure 8, the platform is designed with a clean and professional aesthetic that prioritizes high usability and a layout that is easy to understand. Moreover, the interface features a prominent blue header that clearly communicates the tool's purpose, verifying the credibility of newspaper websites and online articles to the user. In addition, the fake news detector part contains highlighted components such as a simple input field labeled "Enter Website URL" and initiates a deep scan with a single click on the "Analyze" button, allowing users to quickly paste a link and detect. Thus, the following description provides a detailed description of each specific component within this interface:

- Navigation Tabs: Located at the top right, allowing users to switch seamlessly between the "Fake News Detector" and "Spam & Malware" services.

- Input Field & Action Button: A centered, streamlined area designed for efficiency, where the "Analyze" button serves as the primary call-to-action to trigger the backend sequence.

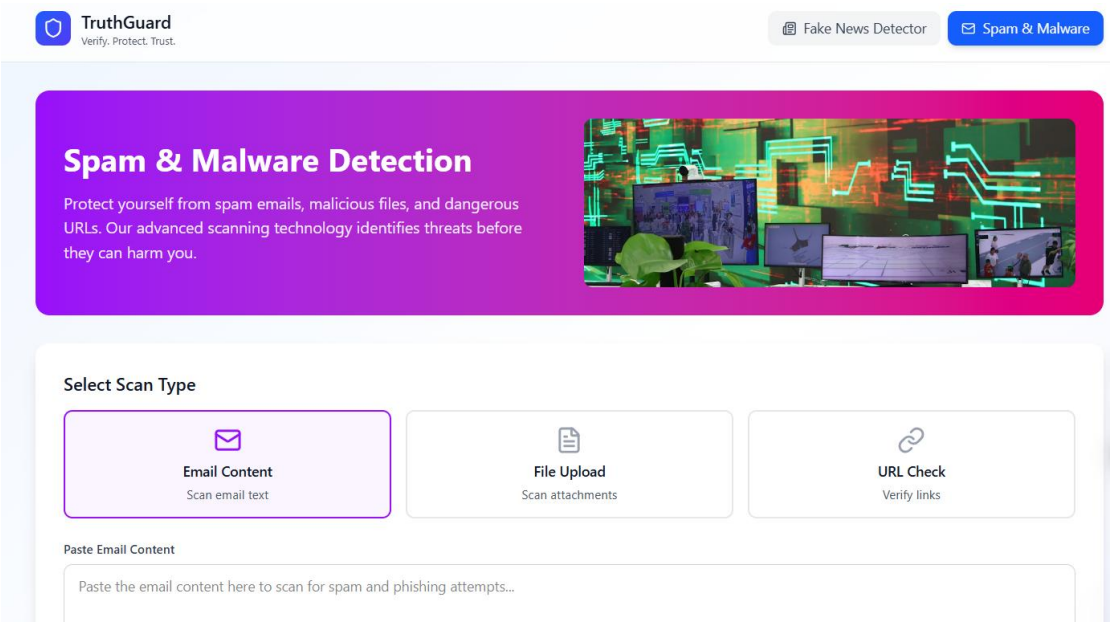

Figure 8: Spam and Malware Mail Detector Interface

Next, building on the established layout, the second major module of our platform is the Spam and Malware Detection space. Additionally, the second interface maintains the project's easy-to-understand philosophy by utilizing a consistent structure that simplifies the process of identifying digital threats. Besides, the following description features the core functional areas of this specific module:

- Service Navigation: The interface uses a distinct purple-themed banner to differentiate it from the news detector, clearly stating its role in protecting users from spam emails, malicious files, and dangerous URLs.
- Scan Type Selection: To ensure high usability, three dedicated interaction cards allow the user to choose their specific detection need: "Email Content," "File Upload," or "URL Check."
- Dynamic Input Area: There are various options for users to choose, including Email Content, File Upload, or URL Check that allow the interface to adapt its input fields to the user's specific needs. Consequently, the layout remains easy to understand by only showing relevant fields, thereby reducing cognitive load and simplifying the data submission process for the FastAPI Backend.
- Action-Oriented Workflow: The layout is designed to guide the user from left-to-right, ensuring the transition from selecting a scan type to viewing the final security report is intuitive and efficient.

Thus, the project design successfully integrates a modular backend with a user-centric interface to ensure the system is both robust and easy to understand. In addition, the team effectively monitors system performance and maintains high-

quality standards by generating sequence diagrams and UI/UX drawings. Notably, these available materials not only guide the development process but also provide a clear roadmap for ensuring the platform remains responsive and reliable for all users.

Part C: Conclusion

In conclusion, the fascinating project has successfully navigated from the initial Project Management Plan to a comprehensive Project Design, ensuring every step in the project keep in the right track. By generating clear management documents, including WBS, Gantt chart, and so on, team members are enabled to monitor progress effectively and ensure the system meets all specified quality and usability requirements.

In addition, the project design plays an indispensable part in the first assignment, which bridges the gap between abstract planning and technical execution. Moreover, my team believes that creating detailed sequence diagrams and UI/UX layouts ensures the code and system architecture align with our team's expected outcomes. Finally, this comprehensive preparation generates a solid foundation for the upcoming development phases and assignments that TruthGuard evolves into a reliable, efficient, and user-friendly solution for the digital community

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